

Dibya Deepta Mishra

Chicago, IL | 650-215-1857 | dibyadeeptamishra@gmail.com | mishradibya.com | [LinkedIn](#)

PhD economist and MS computer scientist studying AI incentives, evaluation, economic value, and the policy effects of technology. Research combines structural econometrics, causal inference, and computational methods.

RESEARCH INTERESTS

Frontier AI governance and firm incentives; multi-agent risks and oversight; AI adoption and productivity; industrial organization; causal policy evaluation; structural econometrics.

EDUCATION

Rice University — PhD, Economics, 2024 (industrial organization, structural econometrics)

Georgia Institute of Technology — MS, Computer Science, 2024 (machine learning and AI)

London School of Economics — MSc, Econometrics and Mathematical Economics, 2017

Indian Institute of Management Calcutta — PGDM/MBA, 2014 | IIT Jodhpur — BTech, Electrical Engineering, 2012

AI POLICY AND CURRENT RESEARCH

SPAR Research Fellow — The Economic Value of AI Capability Gains (voluntary) 2026–present

- Study when marginal model capability gains produce task-level productivity and economic value, accounting for adoption thresholds, verification costs, and complementary bottlenecks.

Dynamic Competition and Frontier AI Model Release (work in progress) 2026

- Developing a model of firms' release timing, capability gains, safety investment, and competition; examining how targeted safety requirements affect release incentives.

AI Agent Swarms and Collective Risk (joint work in progress) 2026

- Analyzing how coordinated less-capable agents share procedures, amplify capability, and propagate failures; examining independent verification before wider adoption.

You Can't Reward What You Can't See — AlignmentxEcon blog series, Part 1 2026

- Connects principal-agent theory to AI oversight and the risk that optimization against a monitor can make its signals less informative.

Do Language Models Represent Utility-Maximizing Choice? (independent research) 2026

- Used layerwise probes and activation-based tests in PyTorch to study decision-relevant signals in open language models.

PROFESSIONAL EXPERIENCE

Cornerstone Research — Associate, Chicago, IL Jan 2025–Sep 2026

- Managed teams of 3–5 analysts processing healthcare claims and proprietary technology-platform datasets with 1M+ records; reviewed data quality, reproducible code, and quantitative work supporting expert testimony.
- Developed causal and structural analyses to test opposing experts' economic theories, identification assumptions, and causal claims; presented methods and limitations to the testifying expert and senior teams.
- Assessed generative AI benchmark reliability and robustness; previously Summer Associate in 2024.

Amazon Retail Pricing — Economist Intern, Seattle, WA May–Aug 2023

- Built demand and elasticity models on 1M+ retail observations using product-text language-model embeddings; improved demand prediction by 5% over baseline and connected estimates to pricing optimization.

Indian School of Business — Research Associate, Hyderabad, India Sep 2017–Apr 2019

- Evaluated policy and market interventions with causal inference on large observational datasets; coauthored a Harvard Business Publishing case study on healthcare incentives.

HSBC — Business and Regulatory Risk Research, Mumbai, India May 2014–Mar 2015

- Analyzed regulatory and macroeconomic developments and presented risk assessments to senior management.

PUBLICATIONS

- AI-assisted teams outperform AI-led teams but not human-only teams in assessing research reproducibility in quantitative social science** *PNAS, 2026*
With Abel Brodeur et al. Collaborative meta-study of human and AI teams assessing research reproducibility.
- Local Population Characteristics and Access Equity of 340B Contract Pharmacies** *2025*
With Maura Coughlin. Health Affairs Scholar 3(7).
- The Wizard of OZ (Opportunity Zones): Spatial Spillovers in Place-Based Programs** *2025*
With Robin Sickles and Yanfei Sun. In Seven Decades of Econometrics and Beyond (Springer).
- Incentives in the Healthcare System** *2019*
With Sisir Debnath and Tarun Jain. Harvard Business Publishing case study.

SELECTED WORKING PAPERS AND POLICY RESEARCH

- Beyond a Free Lunch: Evaluating the Trade-Offs of an Out-of-School Nutrition Program on Adolescent Girls' Human Capital** *JEBO R&R*
With Tanmay Devi, Ajinkya Keskar, and Ritika Sethi. Evaluates how an in-kind transfer outside school affected girls' education and later health in India.
- Pennies for Pads: Subsidized Menstrual Health Products and Adoption in India** *under review*
With Ritika Sethi. Doubly robust difference in differences using staggered subsidized pharmacy rollout and linked survey, administrative, and geospatial data for nearly 500,000 women.
- Quality Considerations and Universal Healthcare** *working paper*
Structural hospital-choice model under limited consideration; machine learning for risk-adjusted quality measures and counterfactuals for information and hospital participation policies.
- Spectrum Allocation to Public Operators: Set-Asides versus Subsidies in Indian Telecom Auctions**
With Rajarshi Bhowal and Nurdaulet Menglibayev. Research in progress.
- Who Matches with Whom: Mergers and Acquisitions in Health Systems** *research in progress*
- Allocating Bundles: An Operations Research Framework** *research in progress*
With Jeremy Fox.

ACADEMIC EXPERIENCE AND SERVICE

Research assistantships: Jeremy T. Fox, Maura Coughlin, and Mallesh Pai (Rice); Tarun Jain and Sisir Debnath (Indian School of Business).

Teaching: Graduate TA at Rice (PhD Microeconomics II, Econometrics I, Math Camp, Industrial Organization, Macroeconomics II); TA at LSE and Indian School of Business.

Professional service: Referee for PLOS ONE and Economic Modelling.

SELECTED PRESENTATIONS AND HONORS

Presentations: Texas Camp Econometrics; DevPEC; ASHEcon; UT Austin Development Economics Workshop; Econometric Society North American and Asia Meetings; RES/SES; SEA; APPAM.

Honors: George R. Zodrow Award (Rice); Inlaks Scholar; Charpak Scholar; Viterbi India Fellow; NTSE Scholar; Rice dissertation research grants.

METHODS AND SOFTWARE

Structural discrete choice and demand estimation; difference in differences, IV, matching, synthetic control; experimental design; AI evaluation and interpretability; Python, PyTorch, Julia, R, Stata, SQL, Git.